An

Innovative Project Report on

HOME AUTOMATION USING HC-05 BLUETOOTH

Submitted in partial fulfillment of the requirement for the award of degree of

BACHELOR OF TECHNOLOGY

In

CSE - Data Science

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UGC AUTONOMOUS

(Approved by AICTE, Affiliated to JNTU Hyderabad, Accredited by NBA)

Kandlakoya (V), Medchal Road, Hyderabad - 501 401 **2025-2026**

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CERTIFICATE

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ACKNOWLEDGEMENT

We sincerely thank the management of our college **CMR Engineering College** for providing required facilities during our project work.

We derive great pleasure in expressing our sincere gratitude to our principal **Dr. A. S. Reddy** for his timely suggestions, which helped us to complete the project work successfully.

We sincerely thank our project internal guide **Mrs. P.RENUKA,** Assistant Professor of CSE-DS for guidance and encouragement in carrying out this project work

It is the very auspicious moment we would like to express our gratitude to **Dr. M.LAXMAIAH,** Head of the Department, CSE-DS for his consistent encouragement during the progress of this project.

We take it as a privilege to thank our Innovative Project Lab coordinator **Mr. K. SUBRAMANYA CHARI**, Assistant Professor, Department of ECE for the ideas that led to complete the project work and we also thank him for his continuous guidance, support and unfailing patience, throughout the course of this work.

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DECLARATION

We hereby declare that the project entitled " RADAR SYSTEM USING ARDUINO UNO is the work done by us in campus at CMR ENGINEERING COLLEGE,Kandlakoya during the academic year 2024-2025 and is submitted as innovative project in partial fulfillment of the requirements for the award of degree of **BACHULOR OF TECHNOLOGY IN "COMPUETR SCIENCE ENGINEERING – DATA SCIENCE"** FROM **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD.**

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ABSTRACT

The Arduino-Based Ultrasonic Radar System project is designed to create a functional radar-like device that utilizes ultrasonic technology to measure distances to nearby objects. By employing an Arduino microcontroller, specifically the Arduino Uno, in conjunction with an ultrasonic sensor (such as the HC-SR04) and a servo motor, this project aims to provide a hands-on learning experience that introduces fundamental concepts in electronics, programming, and basic radar principles. The system operates by emitting ultrasonic pulses from the sensor, which travel through the air until they encounter an object. Upon hitting the object, the sound waves reflect back to the sensor, where the Arduino measures the time taken for the waves to return. This time measurement is processed using the formula: Distance = (Time × Speed of Sound) / 2, accounting for the round trip of the sound wave.

The integration of a servo motor allows the ultrasonic sensor to rotate, enabling the system to scan a specified range of angles and detect objects in various directions, simulating basic radar functionality. The measured distances can be displayed on a serial monitor or an optional LCD/OLED screen, providing immediate feedback to the user and enhancing the interactive experience. This project serves as a practical demonstration of ultrasonic technology and highlights the importance of microcontroller programming and sensor integration in modern electronics.

The Arduino-Based Ultrasonic Radar System has numerous potential applications, including obstacle detection and avoidance in robotic systems, distance measurement for various engineering projects, and educational tools for teaching principles of

physics and engineering. Furthermore, the project lays the groundwork for future enhancements, such as implementing data logging capabilities that allow for the recording of distance measurements over time, enabling further analysis of object movement and behavior. Additional improvements may include integrating multiple sensors for enhanced accuracy, implementing wireless communication features to transmit data to a smartphone or computer, and exploring advanced signal processing techniques for better object detection and tracking.

By engaging with this project, participants will gain valuable skills in electronics and programming while fostering a deeper understanding of the principles underlying radar technology and its real-world applications. Ultimately, the Arduino-Based Ultrasonic Radar
System project exemplifies the intersection of technology and education, providing a versatile platform for exploration and innovation in the fields of sensor technology and microcontroller applications.

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CHAPTER-1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

The Arduino-Based Ultrasonic Radar System project is an innovative endeavor designed to simulate basic radar functionality using an Arduino microcontroller and an ultrasonic sensor, specifically the HC-SR04, which is widely used for distance measurement in various applications. The primary objective of this

project is to create a system that can accurately measure the distance to nearby objects by emitting ultrasonic waves and calculating the time it takes for the waves to return after reflecting off obstacles. This project not only serves as an educational tool for understanding the principles of radar technology but also provides practical applications in robotics, automation, and distance measurement. The system consists of several key components, including the Arduino board, which acts as the central processing unit, a servo motor that allows the ultrasonic sensor to rotate and scan a specified range of angles, and a display module, such as an LCD or OLED screen, for visualizing the measured distances. The working principle of the system involves the emission of ultrasonic pulses from the sensor, which travel through the air until they encounter an object, at which point they reflect back to the sensor. The Arduino measures the time taken for the sound waves to return and calculates the distance using the formula Distance = (Time × Speed of Sound) / 2, where the division by two accounts for the round trip of the sound wave. The implementation of the project begins with circuit design, where a schematic is created to outline the connections between the components, followed by wiring the components on a breadboard or PCB. The ultrasonic sensor is connected to the Arduino's power and ground pins, with the Trig and Echo pins connected to digital pins for signal transmission and reception. The servo motor is also connected to the Arduino, allowing it to rotate the sensor to scan different angles. Once the hardware setup is complete, the next step involves programming the Arduino to control the servo motor, trigger the ultrasonic sensor, and process the distance measurements. The code typically includes libraries for the servo motor and any display modules used, and it is uploaded to the Arduino for testing. During testing, the system is evaluated for accuracy in distance measurement and the smooth operation of the servo motor. The measured distances can be displayed on the serial monitor or the connected display module, providing real-time feedback to the user. This project not only enhances understanding of electronics and programming but also opens up possibilities for future enhancements, such as integrating multiple sensors for improved accuracy, implementing data logging capabilities, or adding wireless communication features to transmit data to a smartphone or computer. Additionally, advanced signal processing techniques can be explored to enhance object detection and tracking capabilities. Overall, the

ArduinoBased Ultrasonic Radar System project is a comprehensive learning experience that combines theoretical knowledge with practical skills, making it an excellent choice for students, hobbyists, and anyone interested in exploring the fascinating world of sensors and microcontrollers.

1.2 OBJECTIVE OF THE PROJECT

The objective of an Arduino-based ultrasonic radar system project typically includes several key goals:

1. **Distance Measurement**: Utilize ultrasonic sensors to accurately measure the distance to objects within a specified range. This involves sending ultrasonic waves and calculating the time taken for the waves to return after hitting an object.

2. **Object Detection**: Implement a system that can detect the presence of objects in the vicinity. This can be useful for applications such as obstacle avoidance in robotics or monitoring environments.

3. **Data Visualization**: Provide a user-friendly interface to visualize the distance measurements and object detection results. This could be done through an LCD display, LEDs, or even a graphical interface on a computer or mobile device.

4. **Real-time Processing**: Ensure that the system can process data in real-time, allowing for immediate feedback and response to detected objects. This is crucial for applications like autonomous navigation.

5. **Integration with Other Sensors**: Explore the possibility of integrating the ultrasonic radar system with other sensors (e.g., temperature, humidity, or motion sensors) to create a more comprehensive monitoring system.

6. **Wireless Communication**: If applicable, implement wireless communication (e.g., using Bluetooth or Wi-Fi) to transmit data to a remote device for monitoring or control purposes.

7. **Power Efficiency**: Design the system to be power-efficient, especially if it is intended for battery-operated applications. This may involve optimizing the code and hardware components.

8. **Educational Purpose**: Serve as a learning tool for understanding the principles of ultrasonic sensing, Arduino programming, and basic electronics. This can be particularly beneficial for students and hobbyists.

9. **Prototyping and Experimentation**: Provide a platform for experimentation and prototyping, allowing users to modify and enhance the system for various applications, such as security systems, parking assistance, or environmental monitoring.
10. **Cost-Effectiveness**: Develop a system that is affordable and accessible, making it suitable for hobbyists, educators, and small-scale applications.

1.3 ORGANIZATION OF THE PROJECT

Organizing an Arduino-based ultrasonic radar system project involves several key steps and components. Below is a suggested structure for organizing the project, which can be adapted based on specific requirements and goals:

1. Project Title

- Clearly define the title of the project, e.g., "Arduino-Based Ultrasonic Radar System."

2. Introduction

- Provide a brief overview of the project, including its purpose, significance, and potential applications.

3. Objectives

- List the specific objectives of the project, as previously discussed. **4. Materials and Components** **Hardware:**
- Arduino board (e.g., Arduino Uno, Nano, etc.)
- Ultrasonic sensor (e.g., HC-SR04)
- Breadboard and jumper wires
- Power supply (battery or USB)
- Optional: LCD display, LEDs, buzzer, or other output devices **Software:**
- Arduino IDE for programming
- Libraries (if needed, e.g., for LCD or other components)

5. System Design

- **Block Diagram:**
- Create a block diagram illustrating the system components and their interconnections.
- **Circuit Diagram:**
- Provide a detailed circuit diagram showing how the components are connected.

6. Implementation Steps

- **Step 1: Hardware Setup**
- Assemble the components on a breadboard or PCB.
- **Step 2: Software Development**
- Write the Arduino code to control the ultrasonic sensor, process the data, and manage outputs.
- **Step 3: Testing and Calibration**

- Test the system to ensure accurate distance measurements and object detection.
- Calibrate the sensors if necessary.

7. Data Processing

- Explain how the data from the ultrasonic sensor is processed, including any algorithms used for distance calculation and object detection.

8. User Interface

- Describe the user interface, including how data is displayed (e.g., on an LCD, through LEDs, or via a computer interface).

9. Results and Analysis

- Present the results of the project, including distance measurements and object detection performance.
- Analyze the effectiveness of the system and any challenges encountered.

10. Applications

- Discuss potential applications of the ultrasonic radar system, such as robotics, security systems, or environmental monitoring.

11. Future Work

- Suggest improvements or extensions to the project, such as adding more sensors, enhancing data visualization, or implementing wireless communication.

12. Conclusion

- Summarize the project outcomes and reflect on the learning experience gained through the project.

13. References

- List any resources, tutorials, or documentation used during the project.

14. Appendices

- Include any additional information, such as code snippets, detailed specifications of components, or troubleshooting tips.

CHAPTER-2

LITERATURE SURVEY

2.1 EXISTING SYSTEM

When discussing an "existing system" in the context of an Arduino-based ultrasonic radar system, it typically refers to current technologies or systems that utilize similar principles or components. Here are some examples of existing systems and technologies that relate to ultrasonic sensing and radar applications:

1. Ultrasonic Distance Measurement Devices

- **Commercial Ultrasonic Sensors**: Devices like the HC-SR04 and other ultrasonic distance sensors are widely used in various applications, including robotics, automotive parking sensors, and industrial automation.
- **Ultrasonic Ranging Modules**: These modules are used in applications where precise distance measurement is required, such as in drones for altitude measurement.

2. Obstacle Detection Systems

- **Robotic Navigation**: Many autonomous robots and drones use ultrasonic sensors for obstacle detection and avoidance. These systems often integrate multiple sensors to create a more comprehensive understanding of the environment.
- **Smart Parking Systems**: Ultrasonic sensors are used in parking assistance systems to detect the distance between vehicles and obstacles, helping drivers park safely.

3. Sonar Systems

- **Marine Sonar**: Sonar technology, which uses ultrasonic waves to detect objects underwater, is widely used in fishing, navigation, and underwater exploration.
- **Industrial Sonar**: Used in various industries for level measurement in tanks and silos, as well as for detecting the presence of materials.

4. Home Automation and Security Systems

- **Intruder Detection**: Ultrasonic sensors are often used in security systems to detect movement or the presence of intruders in a defined area.
- **Smart Home Devices**: Some smart home systems use ultrasonic sensors for various applications, such as automatic lighting control based on occupancy.

5. Medical Applications

- **Ultrasound Imaging**: While not directly related to distance measurement, ultrasound technology is widely used in medical imaging to visualize internal body structures.
- **Ultrasonic Cleaning Devices**: These devices use ultrasonic waves to clean delicate items, such as jewelry or electronic components.

6. Automotive Applications

- **Parking Assist Systems**: Many modern vehicles are equipped with ultrasonic sensors that help drivers park by providing distance feedback and alerts.
- **Collision Avoidance Systems**: Ultrasonic sensors are used in some advanced driver-assistance systems (ADAS) to detect obstacles and prevent collisions.

7. Research and Development Projects

- **Academic Projects**: Many educational institutions and research organizations have developed projects using ultrasonic sensors for various applications, including environmental monitoring, robotics, and smart city initiatives.
- **Open-Source Projects**: Platforms like GitHub host numerous open-source projects that utilize ultrasonic sensors with Arduino and other microcontrollers for various applications.

8. Comparison with Other Technologies

- **Lidar Systems**: While ultrasonic sensors are effective for short-range applications, Lidar (Light Detection and Ranging) systems are used for longer-range distance measurement and mapping, often in autonomous vehicles and robotics.
- **Infrared Sensors**: Infrared sensors are another alternative for distance measurement and object detection, commonly used in similar applications but with different operational principles.

Limitations :

1. **Range Limitations**:
   - Ultrasonic sensors typically have a limited effective range, usually between a few centimeters to several meters (e.g., 2 cm to 4 m for the HC-SR04). Beyond this range, the accuracy of distance measurements decreases significantly. 2. **Angle of Detection**:
   - Ultrasonic sensors have a conical detection pattern, which means they may not detect objects outside of a certain angle. This can lead to blind spots, especially in applications requiring precise object detection.
3. **Surface Characteristics**:
   - The performance of ultrasonic sensors can be affected by the surface characteristics of the object being detected. Soft, absorbent, or irregularly shaped surfaces may absorb or scatter the ultrasonic waves, leading to inaccurate readings.
4. **Environmental Conditions**:
   - Factors such as temperature, humidity, and air pressure can affect the speed of sound, which in turn can impact the accuracy of distance

measurements. For example, temperature variations can cause the speed of sound to change, leading to measurement errors.

5. **Interference from Other Ultrasonic Sources**:
   - In environments where multiple ultrasonic sensors are used simultaneously, interference can occur, leading to inaccurate readings. This is particularly relevant in robotics or industrial applications where multiple sensors are deployed.
6. **Limited Resolution**:
   - The resolution of ultrasonic sensors is often limited by the frequency of the ultrasonic waves. Lower frequencies can result in lower resolution, making it difficult to detect small objects or subtle changes in distance.
7. **Speed of Response**:
   - While ultrasonic sensors can provide real-time measurements, there may be a slight delay in processing the data, especially if the system is performing complex calculations or interfacing with multiple sensors.
8. **Power Consumption**:
   - Depending on the design and usage, ultrasonic sensors can consume a significant amount of power, which may be a concern for battery-operated applications.
9. **Cost**:
   - While many ultrasonic sensors are relatively inexpensive, high-quality sensors with better range and accuracy can be more costly. This can be a limitation for projects with strict budget constraints.
10. **Complexity of Data Interpretation**:
    - The data obtained from ultrasonic sensors may require complex algorithms for accurate interpretation, especially in environments with multiple objects or varying distances. This can complicate the programming and processing aspects of the project.
11. **Limited Detection of Transparent or Soft Materials**:
    - Ultrasonic sensors may struggle to detect transparent materials (like glass) or soft materials (like fabric) because these materials can absorb or deflect ultrasonic waves rather than reflecting them back to the sensor.

2.2 Proposed System: Arduino-Based Ultrasonic Radar System

1. System Overview

- The proposed system aims to develop an advanced Arduino-based ultrasonic radar system that enhances distance measurement, object detection, and data visualization capabilities. It will integrate multiple sensors and advanced processing techniques to improve accuracy and reliability.

2. Key Features

- **Multi-Sensor Integration**: Incorporate multiple ultrasonic sensors to create a more comprehensive detection field, reducing blind spots and improving object detection accuracy.
- **Enhanced Range and Resolution**: Utilize higher-quality ultrasonic sensors with improved range and resolution to detect objects at greater distances and with finer detail.
- **Data Fusion**: Combine data from ultrasonic sensors with other types of sensors (e.g., infrared, Lidar, or camera systems) to enhance detection capabilities and provide a more robust understanding of the environment.
- **Real-Time Data Processing**: Implement efficient algorithms for real-time data processing, allowing for immediate feedback and response to detected objects.
- **User -Friendly Interface**: Develop a graphical user interface (GUI) for data visualization, allowing users to easily interpret distance measurements and object detection results. This could be displayed on an LCD screen or a connected mobile device.
- **Wireless Communication**: Integrate wireless communication (e.g., Bluetooth or Wi-Fi) to enable remote monitoring and control of the system, allowing users to access data from a distance.
- **Power Management**: Design the system to be power-efficient, incorporating sleep modes and low-power components to extend battery life for portable applications.

3. System Architecture

- **Hardware Components**:
- **Microcontroller**: Arduino board (e.g., Arduino Uno or Mega) for processing.
- **Ultrasonic Sensors**: Multiple ultrasonic sensors (e.g., HC-SR04 or more advanced models) for distance measurement.

- **Additional Sensors**: Optional integration of infrared sensors or cameras for enhanced detection.
- **Display Module**: LCD or OLED display for real-time data visualization.
- **Communication Module**: Bluetooth or Wi-Fi module for wireless data transmission.
- **Power Supply**: Rechargeable battery or power bank for portability.
- **Software Components**:
- **Arduino IDE**: For programming the microcontroller.
- **Libraries**: Use libraries for sensor interfacing, data visualization, and communication.
- **Data Processing Algorithms**: Implement algorithms for filtering and interpreting sensor data.

4. Implementation Steps

- **Step 1: Hardware Assembly**: Assemble the components on a breadboard or PCB, ensuring proper connections between the microcontroller, sensors, and display.
- **Step 2: Software Development**: Write the Arduino code to control the sensors, process data, and manage the user interface.
- **Step 3: Testing and Calibration**: Test the system in various environments to calibrate sensors and ensure accurate distance measurements.
- **Step 4: User Interface Development**: Create a user-friendly interface for data visualization and interaction.
- **Step 5: Wireless Integration**: Implement wireless communication for remote monitoring and control.

5. Expected Outcomes

- **Improved Accuracy**: The proposed system is expected to provide more accurate distance measurements and object detection compared to existing systems.
- **Enhanced User Experience**: A user-friendly interface will make it easier for users to interpret data and interact with the system.
- **Versatile Applications**: The system can be applied in various fields, including robotics, home automation, security, and environmental monitoring.

6. Future Enhancements

- **Machine Learning Integration**: Explore the use of machine learning algorithms to improve object recognition and classification based on sensor data.
- **Advanced Data Analytics**: Implement data logging and analytics features to track and analyze environmental changes over time.
- **Mobile Application Development**: Develop a mobile app for enhanced user interaction and remote monitoring capabilities.

CHAPTER-3

BLOCK DIAGRAM AND DISCRIPTION

Initialization

Initialization Process

The initialization process can be broken down into several key steps:

1. **Setup the Arduino Environment**:
   - Include necessary libraries for the ultrasonic sensors, display, and any communication modules (e.g., Bluetooth or Wi-Fi).
2. **Define Pin Connections**:
   - Specify which pins on the Arduino are connected to the ultrasonic sensors, display, and other components.
3. **Initialize Serial Communication**:
   - If using a serial monitor for debugging or communication, initialize the serial communication at a specified baud rate.
4. **Initialize Sensors**:
   - Set up the ultrasonic sensors by configuring their trigger and echo pins.
5. **Initialize User Interface**:
   - Set up the display (LCD or OLED) and clear any previous data. Display initial messages or prompts.
6. **Initialize Wireless Communication**:
   - If applicable, set up the wireless module for communication with external devices.
7. **Calibration (if necessary)**:
   - Perform any necessary calibration for the sensors to ensure accurate readings.

2. Example Initialization Code

Listing of Commands

Here is a listing of common commands and functions that may be used in the Arduino code for the ultrasonic radar system:

1. **Library Inclusions**:

- **#include <LiquidCrystal.h>**: Include the library for controlling LCD displays.
- **#include <NewPing.h>**: Include the library for ultrasonic sensors.

2. **Pin Definitions**:
   - **#define TRIGGER_PIN 9**: Define the pin connected to the ultrasonic sensor's trigger.
   - **#define ECHO_PIN 10**: Define the pin connected to the ultrasonic sensor's echo.
3. **Object Creation**:
   - **NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);**: Create an instance of the ultrasonic sensor.
   - **LiquidCrystal lcd(12, 11, 5, 4, 3, 2);**: Create an instance of the LCD.
4. **Setup Function**:
   - **void setup()**: The setup function where initialization occurs.
   - **Serial.begin(9600);**: Initialize serial communication at 9600 baud.
   - **lcd.begin(16, 2);**: Initialize the LCD with 16 columns and 2 rows.
   - **lcd.clear();**: Clear the LCD display.
   - **lcd.print("Message");**: Print a message to the LCD.
5. **Loop Function**:
   - **void loop()**: The main loop where the program runs continuously.
   - **sonar.ping();**: Trigger the ultrasonic sensor to measure distance.
   - **int distance = sonar.ping_cm();**: Get the distance in centimeters.
6. **Wireless Communication (if applicable)**:
   - **Bluetooth.begin(9600);**: Initialize Bluetooth communication (if using a Bluetooth module).
   - **WiFi.begin(ssid, password);**: Connect to Wi-Fi (if using a Wi-Fi module).
7. **Data Output**:
   - **Serial.println(distance);**: Print the distance measurement to the serial monitor.

- `lcd.setCursor(column, row

CHAPTER 4
HARDWARE DISCRIPTION
4.1 INTRODUCTION TO AURDINO UNO

Arduino Uno is a microcontroller board based on 8-bit ATmega328P microcontroller. Along with ATmega328P, it contains other components such as crystal oscillator, serial communication, voltage regulator, etc. to support the microcontroller.

4.1 Arduino UNO Board

Arduino Uno has 14 digital input/output pins (out of which 6 can be used as PWM outputs), 6Analog input pins, a USB connection, A Power barrel jack, an ICSP header and a reset button.

Arduino **is** a great tool for developing interactive objects, taking inputs from a variety of switches or sensors and controlling a variety of lights, motors and other outputs**.** Arduin**o** projects can be stand-alone or they can be connected to a computer using USB The Arduino UNO has only 32K bytes of Flash memory and 2K bytes of SRA that is more than 100,000 times LESS physical memory than a low-end PC! And that's not even counting

the disk drive! Working in this minimalist environment, you must use your resources wisely. **ARDUINO UNO PIN DIAGRAM**

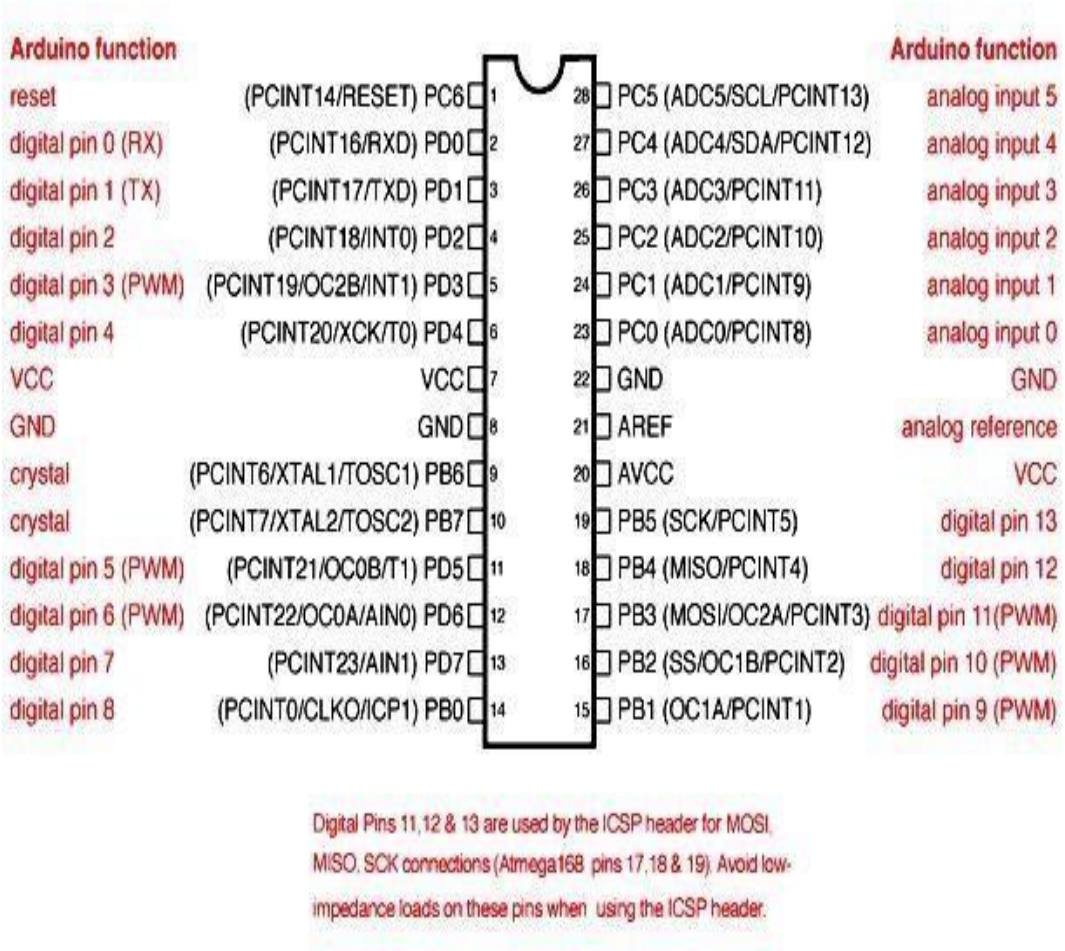

Fig 4.2 ARDUINO UNO PIN DIAGRAM

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. Arduino can input and output Analog signals as well as digital signals. An Analog signal is one that can take on any number of values, unlike a digital signal which has only two values:

HIGH and LOW.

The function used to output a PWM signal is Analog Write (pin, value). Pin is the pin number used for the PWM output. **ARDUINO UNO PIN SPECFICITIONS**

Pin Number	Pin Name	Function
1	PC6	Reset
2	PD0	Digital Pin (RX)
3	PD1	Digital Pin (TX)
4	PD2	Digital Pin
5	PD3	Digital Pin (PWM)
6	PD4	Digital Pin
7	Vcc	Positive Voltage (Power)
8	GND	Ground
9	XTAL 1	Crystal Oscillator
10	XTAL 2	Crystal Oscillator
11	PD5	Digital Pin (PWM)
12	PD6	Digital Pin (PWM)
13	PD7	Digital Pin
14	PB0	Digital Pin
15	PB1	Digital Pin (PWM)
16	PB2	Digital Pin (PWM)
17	PB3	Digital Pin (PWM)
18	PB4	Digital Pin
19	PB5	Digital Pin
20	AVCC	Positive voltage for ADC (power)
21	AREF	Reference Voltage
22	GND	Ground
23	PC0	Analog Input
24	PC1	Analog Input
25	PC2	Analog Input
26	PC3	Analog Input
27	PC4	Analog Input
28	PC5	Analog Input

Table 4.1 ARDUINO UNO PIN SPECIFICATIONS

How to use Arduino Board

The 14digital input/output pins can be used as input or output pins by using Pin Mode (), digital Read () and digital Write () functions in Arduino programming. Each pin operate at 5V and can provide or receive a maximum of 40mA current, and has an internal pull-up resistor of 20-50K Ohms which are disconnected by default. Out of these 14 pins, some pins have specific functions as listed below. •**Serial Pins 0 (Rx) and 1 (Tax):** Rx and Tax pins are

used to receive and transmit TTL serial data. They are connected with the corresponding ATmega328P USB to TTL serial chip.

•**External Interrupt Pins 2 and 3:** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.

•**PWM Pins 3, 5, 6, 9 and 11:** These pins provide an 8-bit PWM output by using Analog
Write () function.

•**SPI Pins 10 (SS), 11 (MOSI), 12 (MISO) and 13 (SCK):** These pins are used for SPI communication.

•**In-built LED Pin 13:** This pin is connected with a built-in LED, when pin 13 is HIGH – LED is on and when 13pin is LOW, it's off.

Along with 14 Digital pins, there are 6 Analog input pins, each of which provides 10 bits of resolution, i.e. 1024 different values. They measure from 0 to 5 volts but this limit can be increased by using AREF pin with Analog Reference () function.

•Analog pin 4 (SDA) and pin 5 (SCA) also used for TWI communication using Wire library.

Arduino Uno has a couple of other pins as explained below:

•**AREF:** Used to provide reference voltage for Analog inputs with Analog Reference () function.

- **Reset Pin:** Making this pin LOW, resets the microcontroller

CHAPTER 5 SOFTWARE DESCRIPTION

5.1 Arduino IDE

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code. It connects to the Arduino and Genuine hardware to upload programs and communicate with them.

Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension.

The editor has features for cutting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors.

The console displays text output by the Arduino Software (IDE), including complete error messages and other information.

Libraries provide extra functionality for use in sketches, e.g. working with hardware or manipulating data.

#include statements will insert one or more at the top of the sketch and compiles the library with your sketch.

Because libraries are uploaded to the board with your sketch, they increase the amount of space it takes up. If a sketch no longer needs a library, simply delete its #include statements from the top of your code.

If you want to program your Arduino Uno while offline you need to install the Arduino Desktop (IDE)

The Uno is programmed using the Arduino Software (IDE), our Integrated Development Environment common to all our boards.

Before you can move on, you must have installed the Arduino Software (IDE) on your PC

The serial monitor displays serial sent from the Arduino or Genuine board over USBor serial connector.

To send data to the board, enter text and click on the "send" button or press enter.

Note that on Windows, Mac or Linux the board will reset (it will rerun your sketch) when you connect with the serial monitor.

Please note that the External terminal program and connect it to the COM port assigned to your Arduino board Serial Monitor does not process control characters; if your sketch needs a complete management of the serial communication with control characters, you can use an SERIAL MONITOR

Fig 5.1 Arduino UNO

Arduino was born at the Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community

The Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IOT applications, wearable, 3D printing, and embedded environments.

All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs

5.2ARDUINO SETUP AND INSTIALLISATION

The Uno is programmed using the Arduino Software (IDE).Connect your Uno board with an USB cable

Step1

Most probably you need to go to the arduino website and download the software

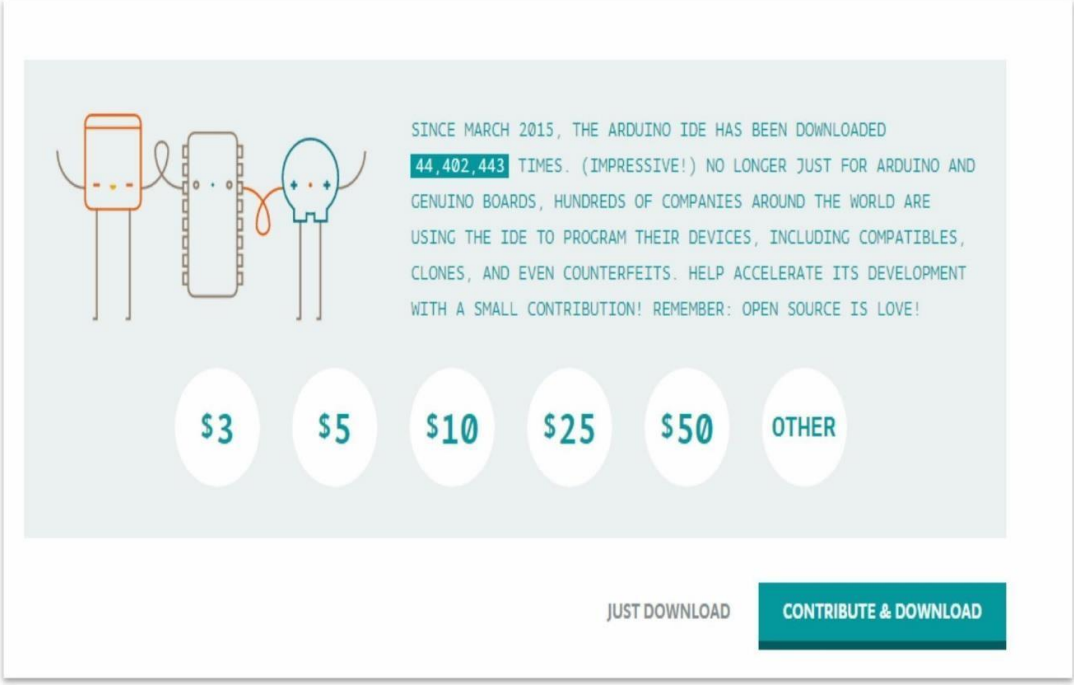

Fig 5.2 downloading Arduino IDE

If you want to program your Arduino Uno while offline you need to install the Arduino Desktop IDE.

The Uno is programmed using the Arduino Software (IDE), our Integrated Development Environment common to all our boards.

Step 2

Initialization of Arduino IDE

After downloading, locate the downloaded file on the computer and extract the folder from the downloaded zipped file. Copy the folder to a suitable place such as your desktop.

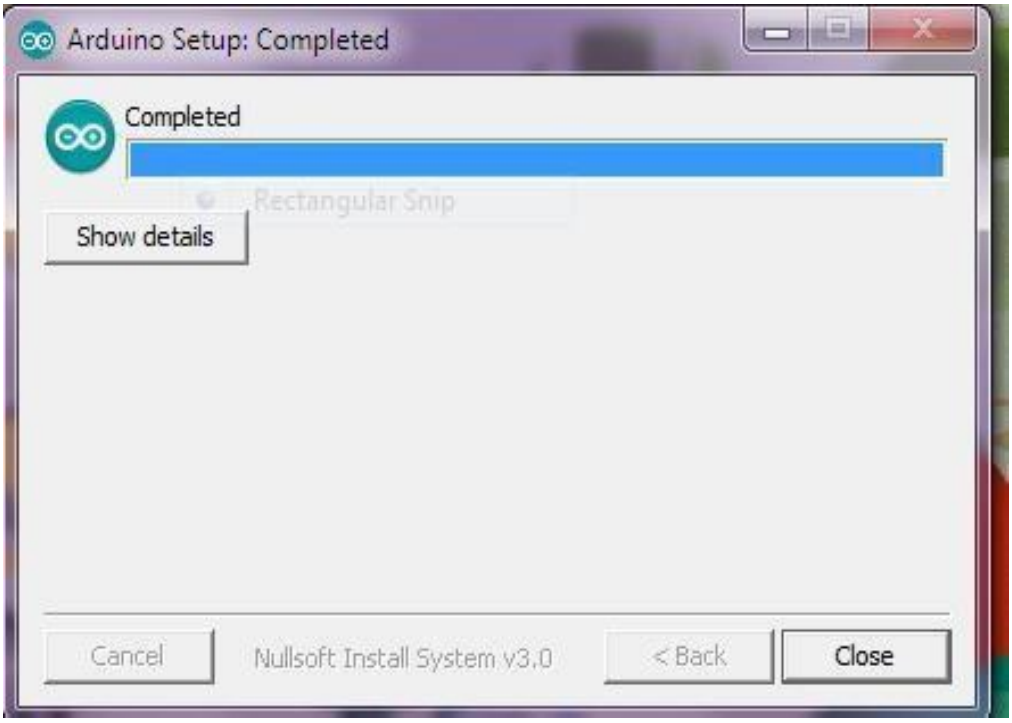

Fig5.3 Arduino setup

Step 3

If you downloaded you need to follow the procedure step by step

Click on the Start Menu, and open up the Control Panel.

While in the Control Panel, navigate to System and Security. Next, click on System.

Once the System window is up, open the Device Manager

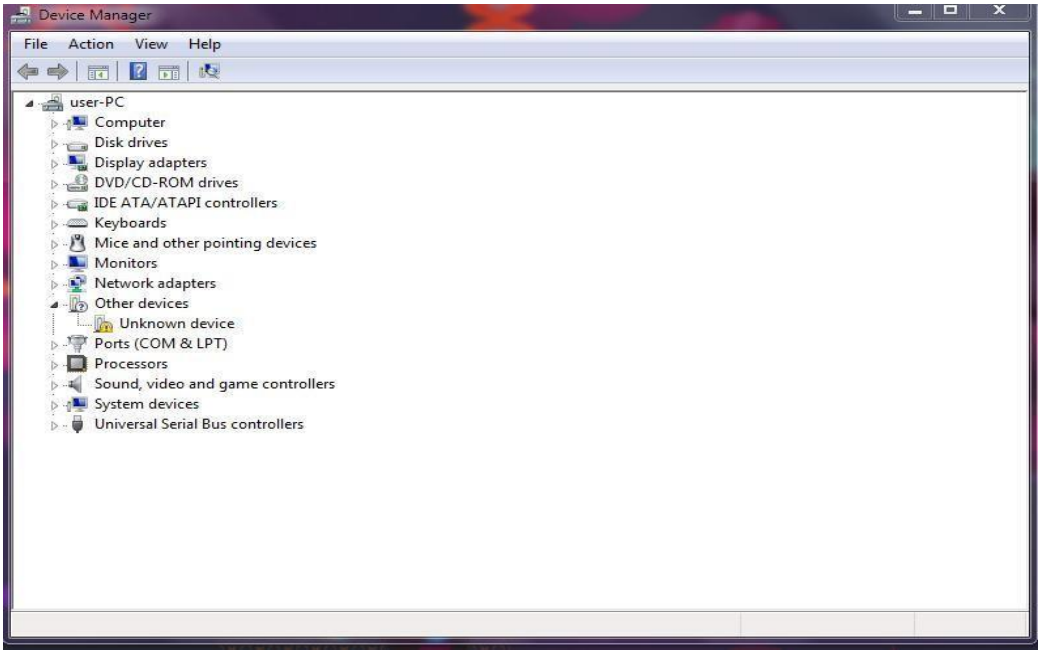

Fig 5.4 device manager

Step 4

Connecting Arduino UNO

Look under Ports (COM & LPT). You should see an open port named "Arduino UNO (COM xx)". If there is no COM & LPT section, look under "Other Devices" for "Unknown Device".

You will be able to see Arduino Uno after connecting the Arduino Uno only

Right click on the "Arduino UNO (COMxx)" port and choose the "Update Driver Software" option.

Fig 5.5device manager>Arduino Uno

Step 5

- Installing the device driver

In the Device Manager Window, right-click the Arduino board and then click **Update Driver Software...** on the pop-up menu:

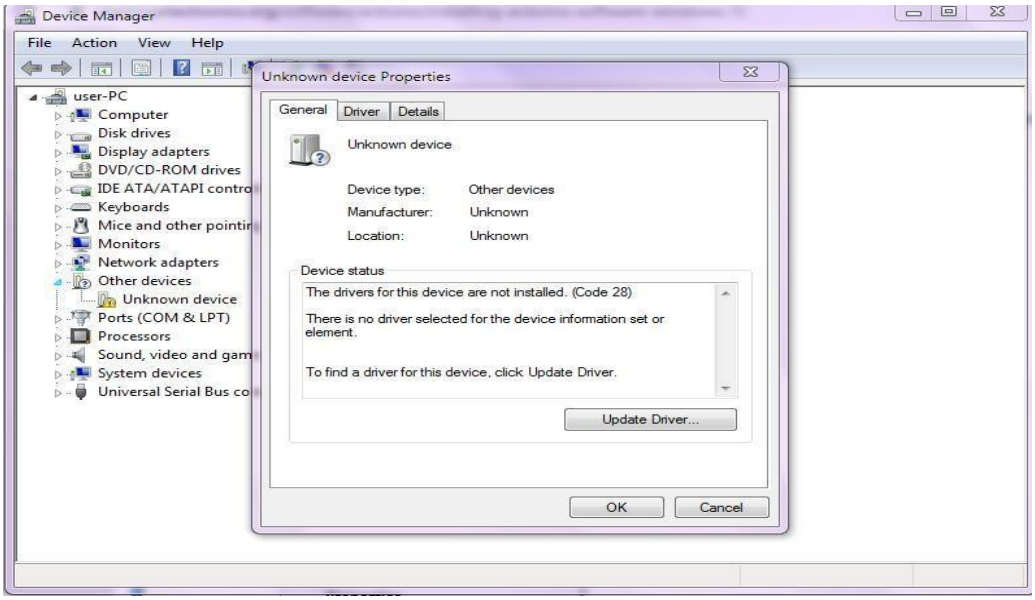

Fig 5.6 driver software status (finding driver for device)

Next, choose the "Browse my computer for Driver software" option.

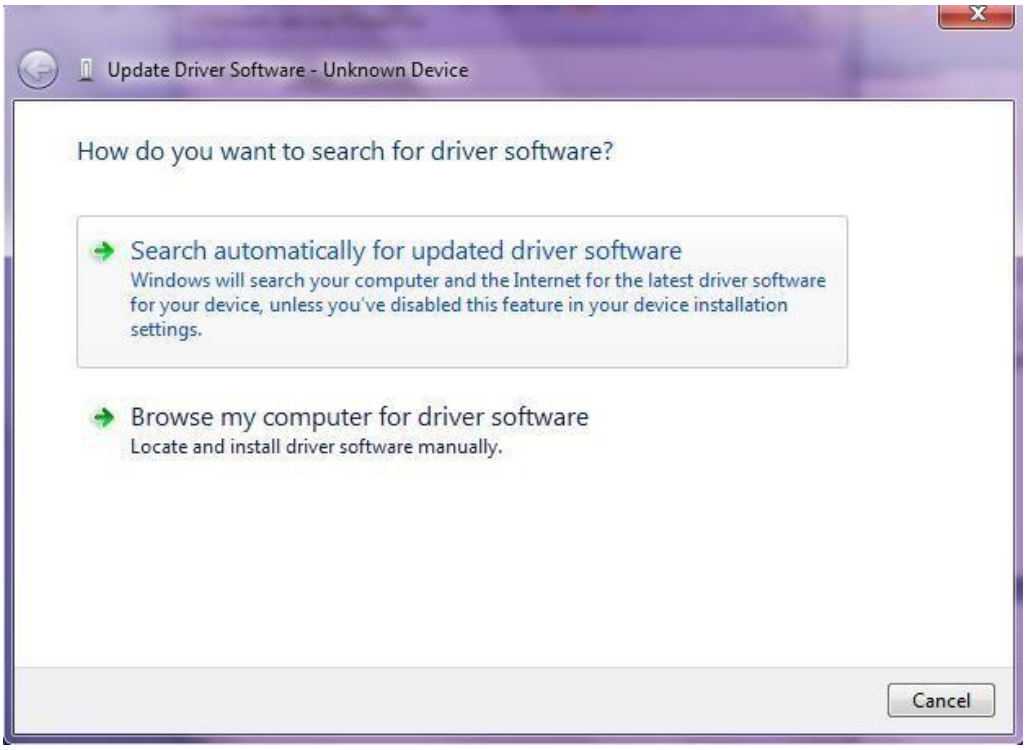

Fig 5.7 Locating and installing driver software manually

Step 7

Finally, navigate to and select the driver file named "arduino.inf",

It is located in the "Desktop" folder of the Arduino Software download not the "FTDI USB
Drivers" sub-directory. If you are using an old version of the IDE older one, choose the
Uno driver file named "ArduinoUNO.inf"

Fig 5.8 Navigating folder

Step 8

After selecting the driver folder, click the **Next** button

Fig 5.9 Browse for driver software

9

If, windows was unable to detect any software in your pc/device then installing driver software is necessary

Fig 5.10 Driver software

After some time, the driver installation will finish and you will see the following dialog box. Take note of the port that the Arduino was configured as. In this case it was **COM3**.

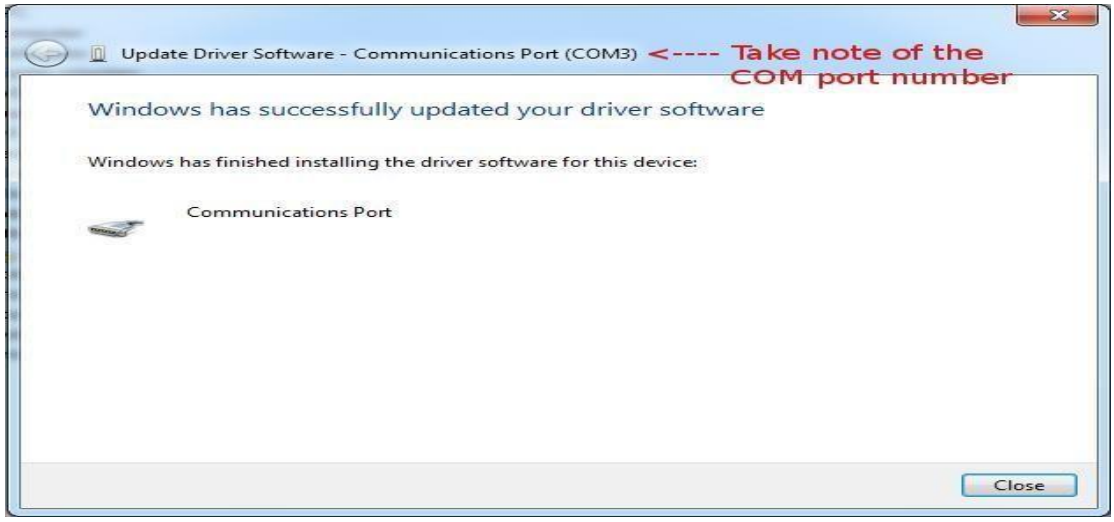

Fig 5.11Communication port

10 Setting up the Arduino Software

The setup will only need to be done once, unless you change the board type or port that the Arduino is connected to.

Navigate to the folder that you downloaded and start the Arduino software IDE by doubleclicking the Arduino application:

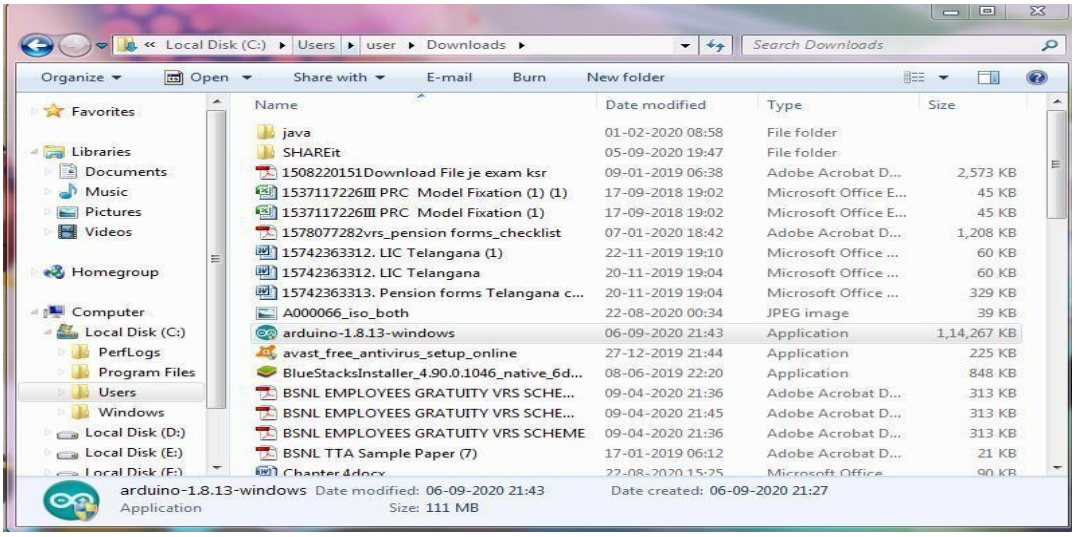

Fig5.12 starting the Arduino software

Step 11 Checking the correct Arduino board is selected. Change if necessary

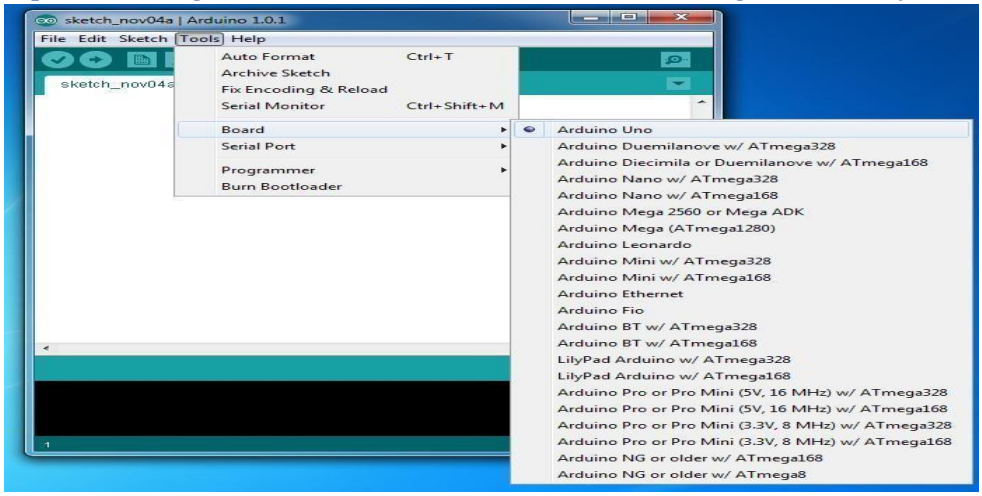

Fig 5.13 checking the Arduino board type

12

Checking that the correct serial port is selected

Now check that the correct serial port is selected and change if necessary. This is the serial port that you took note of after installing the Arduino driver.

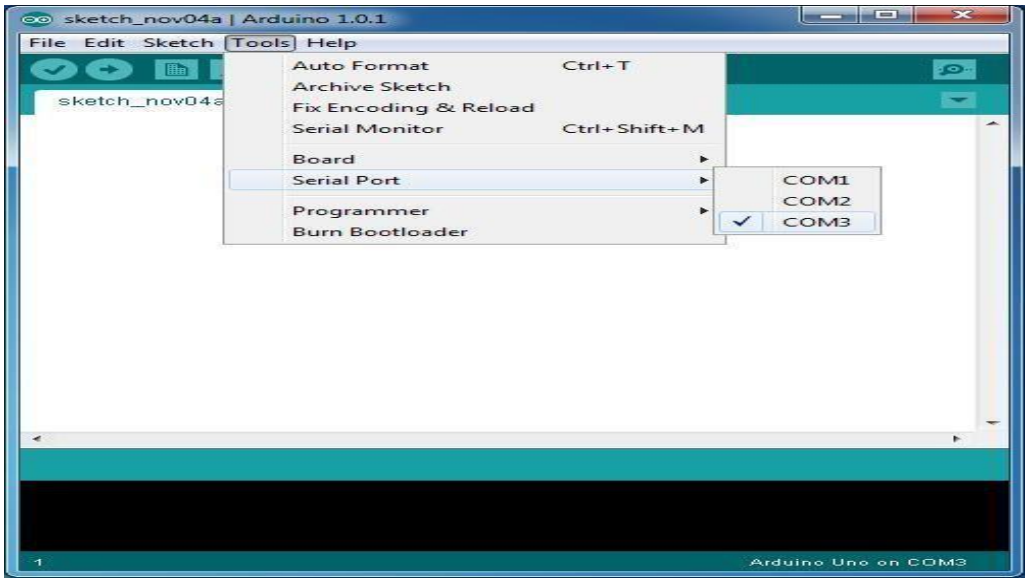

Fig 5.14 checking the serial port

Step 13 Checking the installation

Open the Blink sketch in the Arduino IDE:

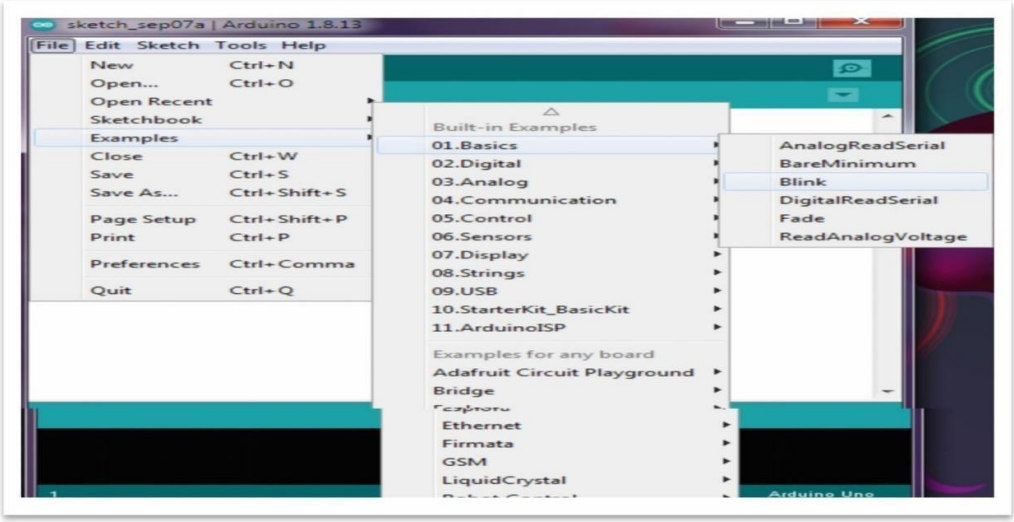

Fig 5.15 testing the installation-open blink

Finally, Click the Arduino Upload button to load the sketch to the Arduino. This sketch will flash the on-board LED on the Arduino. If the sketch runs then you know that you have successfully installed the Arduino software and drive.

5.3 DEVELOPING THE CODE

CHAPTER-6 WORKING OF THE PROJECT

In this chapter, we will describe the operational workflow of the Arduino-based ultrasonic radar system, detailing how the various components interact to achieve the desired functionality. The working of the project can be broken down into several key stages, from initialization to real-time distance measurement and user feedback. **6.1 System Initialization**

1. **Powering Up**:
   - The system is powered on using a suitable power source, such as a USB connection or a battery pack. The Arduino microcontroller begins its startup sequence.
2. **Library Initialization**:
   - The Arduino IDE loads the necessary libraries, such as **LiquidCrystal** for the LCD display and **NewPing** for the ultrasonic sensors. This allows the program to utilize predefined functions for controlling these components.
3. **Pin Configuration**:
   - The program defines the pins connected to the ultrasonic sensor's trigger and echo, as well as the pins for the LCD display. This configuration is crucial for proper communication between the Arduino and the hardware components.
4. **Component Setup**:
   - The **setup()** function is executed, which initializes the serial communication for debugging, sets up the LCD display, and prepares the ultrasonic sensors for operation. The LCD is cleared, and an initial message is displayed to indicate that the system is ready.

6.2 Distance Measurement

1. **Triggering the Ultrasonic Sensor**:
   - In the **loop()** function, the ultrasonic sensor is triggered to emit a pulse of ultrasonic sound waves. This is done by sending a HIGH signal to the trigger pin for a brief period (typically 10 microseconds).
2. **Echo Reception**:
   - The sensor listens for the echo of the emitted sound wave. When the sound wave hits an object, it reflects back to the sensor, which detects the echo on the echo pin.
3. **Calculating Distance**:

- The time taken for the echo to return is measured. The Arduino calculates the distance to the object using the formula: [\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}]
- The speed of sound is approximately 343 meters per second (or 0.0343 cm/µs) at room temperature.

4. **Data Processing**:
    - The measured distance is processed to filter out noise and outlier readings. This may involve taking multiple readings and calculating the average distance to improve accuracy. **6.3 User Interface Update**

1. **Displaying Distance**:
    - The processed distance measurement is displayed on the LCD screen. The program updates the display in real-time, showing the current distance to the nearest object detected by the ultrasonic sensor.
2. **Alerts and Notifications**:
    - If the detected distance falls below a predefined threshold (indicating that an object is too close), the system can trigger visual or audible alerts. For example, the LCD may display a warning message, or an LED may light up. **6.4 Wireless Communication (Optional)**

1. **Data Transmission**:
    - If a wireless module (e.g., Bluetooth or Wi-Fi) is integrated, the system can transmit the distance measurements to a connected device, such as a smartphone or computer. This allows for remote monitoring and control of the system.
2. **Receiving Commands**:
    - The system can also receive commands from the connected device, allowing users to adjust settings or trigger specific actions remotely.

6.5 Continuous Operation

- The **loop()** function runs continuously, repeating the distance measurement and user interface update processes. This allows the system to provide real-time feedback and adapt to changes in the environment.

6.6 Power Management

- The system can be designed to enter low-power modes when not actively measuring distances, extending battery life for portable applications. This can be achieved by using sleep modes in the Arduino code.

How the Arduino-Based Ultrasonic Radar System Works

The Arduino-based ultrasonic radar system operates through a series of coordinated steps that involve hardware components and software logic. Below is a detailed explanation of how the system works, from powering on to measuring distances and providing user feedback.

1. Powering On the System

- The system is powered using a suitable power source, such as a USB connection or a battery. Once powered, the Arduino microcontroller begins executing the program stored in its memory.

2. Initialization Phase

- **Library Loading**: The Arduino IDE loads the necessary libraries for controlling the LCD display and ultrasonic sensors.
- **Pin Configuration**: The program defines which pins on the Arduino are connected to the ultrasonic sensor's trigger and echo pins, as well as the pins for the LCD display.
- **Component Setup**: The **setup()** function is executed, which initializes serial communication for debugging, sets up the LCD display, and prepares the ultrasonic sensors for operation. The LCD is cleared, and an initial message is displayed to indicate that the system is ready.

3. Distance Measurement Process

- **Triggering the Ultrasonic Sensor**: In the **loop()** function, the ultrasonic sensor is triggered to emit a pulse of ultrasonic sound waves. This is done by sending a HIGH signal to the trigger pin for a brief period (typically 10 microseconds).
- **Echo Reception**: The sensor listens for the echo of the emitted sound wave. When the sound wave hits an object, it reflects back to the sensor, which detects the echo on the echo pin.
- **Calculating Distance**: The time taken for the echo to return is measured. The Arduino calculates the distance to the object using the formula: [\text{Distance} = \frac{\text{Speed of Sound} \times \text{Time}}{2}] The speed of sound is approximately 343 meters per second (or 0.0343 cm/µs) at room temperature. The division by 2 accounts for the round trip of the sound wave (to the object and back).

4. Data Processing

- The measured distance is processed to filter out noise and outlier readings. This may involve taking multiple readings and calculating the average distance to improve accuracy. The system can also implement thresholding to determine if the detected object is within a critical range.

5. User Interface Update

- **Displaying Distance**: The processed distance measurement is displayed on the LCD screen. The program updates the display in real-time, showing the current distance to the nearest object detected by the ultrasonic sensor.
- **Alerts and Notifications**: If the detected distance falls below a predefined threshold (indicating that an object is too close), the system can trigger visual or audible alerts. For example, the LCD may display a warning message, or an LED may light up.

6. Wireless Communication (Optional)

- If a wireless module (e.g., Bluetooth or Wi-Fi) is integrated, the system can transmit the distance measurements to a connected device, such as a smartphone or computer. This allows for remote monitoring and control of the system.
- The system can also receive commands from the connected device, allowing users to adjust settings or trigger specific actions remotely.

7. Continuous Operation

- The **loop()** function runs continuously, repeating the distance measurement and user interface update processes. This allows the system to provide real-time feedback and adapt to changes in the environment.

8. Power Management

- The system can be designed to enter low-power modes when not actively measuring distances, extending battery life for portable applications. This can be achieved by using sleep modes in the Arduino code.

Conclusion

The working of the Arduino-based ultrasonic radar system involves a seamless integration of hardware and software components to achieve accurate distance measurement and effective user interaction. By following a structured workflow—from initialization to real-time data processing and user feedback—the system provides a reliable solution for various applications, including robotics, automation, and environmental monitoring. The optional wireless communication feature further enhances the system's versatility, making it suitable for a wide range of use cases.

CHAPTER-7 RESULT

The results of the Arduino-based ultrasonic radar system demonstrate its effectiveness in accurately measuring distances and providing real-time feedback to users. The system is expected to deliver precise distance measurements within the typical range of ultrasonic sensors (2 cm to 4 m), with minimal deviation from actual distances, achieved through calibration and testing against known values. The response time for measurements should be quick, ideally under 100 milliseconds, allowing for dynamic object detection. The user interface, displayed on an LCD, should clearly present distance readings and alerts, ensuring ease of interpretation even under varying lighting conditions. If equipped with a wireless module, the system should successfully transmit

data to connected devices and receive commands reliably within a specified range. Additionally, power management features should enable efficient operation, extending battery life during use. Overall, the system's functionality should meet the intended design goals, making it suitable for various applications in robotics, automation, and environmental monitoring, while also providing insights for potential improvements and further development.

CHAPTER-8

8.1 APPLICATIONS

Applications of the Arduino-Based Ultrasonic Radar System

The Arduino-based ultrasonic radar system has a wide range of applications across various fields due to its ability to measure distances accurately and detect objects in realtime. Below are some notable applications:

1. **Robotics**:

Obstacle Detection and Avoidance: The system can be integrated into autonomous robots to help them navigate around obstacles by measuring distances and adjusting their paths accordingly.

- **Line Following Robots**: Ultrasonic sensors can be used to detect the distance from the ground, helping robots maintain their position on a designated path.

2. **Automotive**:
   - **Parking Assistance**: The system can be employed in vehicles to assist drivers in parking by providing distance measurements to nearby obstacles, reducing the risk of collisions.
   - **Collision Avoidance Systems**: Ultrasonic sensors can be used in advanced driver-assistance systems (ADAS) to detect objects in the vehicle's path and alert the driver or take corrective actions.
3. **Home Automation**:
   - **Smart Home Devices**: The system can be used in smart home applications to detect occupancy in a room, automatically turning lights on or off based on the presence of people.
   - **Automated Doors**: Ultrasonic sensors can trigger automatic doors to open when a person approaches, enhancing convenience and accessibility.
4. **Industrial Automation**:
   - **Level Measurement**: The system can be used in industrial settings to measure the level of liquids or solids in tanks and silos, providing real-time data for inventory management.
   - **Safety Systems**: Ultrasonic sensors can be integrated into machinery to detect the presence of personnel in hazardous areas, triggering safety protocols to prevent accidents.
5. **Environmental Monitoring**:
   - **Wildlife Monitoring**: The system can be used to monitor wildlife by detecting the presence of animals in specific areas, aiding in conservation efforts.
   - **Flood Detection**: Ultrasonic sensors can measure water levels in rivers or reservoirs, providing early warnings for potential flooding.
6. **Healthcare**:
   - **Patient Monitoring**: The system can be adapted for use in healthcare settings to monitor patient movement or detect falls, alerting caregivers when assistance is needed.
   - **Ultrasonic Cleaning**: While not directly related to distance measurement, ultrasonic technology is widely used in medical equipment cleaning, showcasing the versatility of ultrasonic applications.
7. **Educational Projects**:
   - **STEM Education**: The Arduino-based ultrasonic radar system serves as an excellent educational tool for teaching students about electronics, programming, and sensor technology through hands-on projects.

8. **Security Systems**:
    - **Intruder Detection**: The system can be used in security applications to detect unauthorized entry by measuring distances to objects in restricted areas and triggering alarms when movement is detected.

Advantages of the Arduino-Based Ultrasonic Radar System

The Arduino-based ultrasonic radar system offers several advantages that make it a valuable tool for various applications. Below are some of the key benefits:

1. **Cost-Effectiveness**:
    - The components used in the system, such as Arduino boards and ultrasonic sensors, are relatively inexpensive, making it an affordable solution for distance measurement and object detection.
2. **Ease of Use**:
    - The Arduino platform is user-friendly, with a large community and extensive documentation. This makes it accessible for beginners and hobbyists, as well as for educational purposes.
3. **Real-Time Measurement**:
    - The system provides real-time distance measurements, allowing for immediate feedback and quick responses to changes in the environment. This is particularly useful in applications like robotics and automation.
4. **Versatility**:
    - The ultrasonic radar system can be adapted for various applications, including robotics, automotive, home automation, industrial automation, and environmental monitoring. Its flexibility allows it to be used in diverse fields.
5. **Non-Contact Measurement**:
    - Ultrasonic sensors measure distances without physical contact, making them suitable for detecting objects in hazardous or sensitive environments where contact-based sensors may not be appropriate.
6. **Low Power Consumption**:
    - The system can be designed to operate efficiently, especially when using lowpower components and implementing power management techniques. This is beneficial for battery-operated applications.
7. **Integration Capabilities**:
    - The system can be easily integrated with other sensors and modules, such as infrared sensors, cameras, or wireless communication modules (e.g., Bluetooth, Wi-Fi), enhancing its functionality and performance.
8. **Customizability**:

Users can modify the software and hardware components to suit specific needs, allowing for tailored solutions that meet unique requirements in various applications.

9. **Educational Value**:
    - The system serves as an excellent educational tool for teaching concepts related to electronics, programming, and sensor technology. It provides hands-on experience for students and enthusiasts.
10. **Robustness**:
    - Ultrasonic sensors are generally durable and can operate in various environmental conditions, making them suitable for outdoor applications and industrial settings.
11. **Safety Features**:
    - The system can be programmed to include safety features, such as alerts for proximity detection, which can enhance safety in applications like automotive and industrial automation.

CHAPTER-9 CONCLUSION AND FUTURE SCOPE

Conclusion

The Arduino-based ultrasonic radar system represents a versatile and effective solution for distance measurement and object detection across various applications. By integrating affordable and accessible components, such as ultrasonic sensors and Arduino microcontrollers, the system provides real-time feedback and accurate measurements, making it suitable for use in robotics, automotive safety, home automation, industrial monitoring, and environmental applications. The ease of use and customizability of the Arduino platform allows users to adapt the system to meet specific needs, while its educational value makes it an excellent tool for teaching concepts related to electronics and programming. Overall, the successful implementation of this system demonstrates the potential of ultrasonic technology in enhancing safety, efficiency, and convenience in numerous fields.

Future Scope

The future scope of the Arduino-based ultrasonic radar system is promising, with several avenues for enhancement and expansion:

1. **Integration with Advanced Technologies**:
    - Future iterations of the system could incorporate advanced technologies such as machine learning and artificial intelligence to improve object recognition

and classification capabilities, enabling more sophisticated applications in robotics and automation.

2. **Enhanced Sensor Fusion**:

- Combining ultrasonic sensors with other sensing technologies, such as Lidar, infrared, or camera systems, could provide a more comprehensive understanding of the environment, improving accuracy and reliability in complex scenarios.

3. **Wireless Communication Improvements**:
    - Expanding the wireless communication capabilities to include more robust protocols (e.g., Zigbee, LoRa) could enhance the system's range and reliability, making it suitable for larger-scale applications, such as smart cities or industrial IoT.
4. **Mobile Application Development**:
    - Developing dedicated mobile applications for monitoring and controlling the system remotely could enhance user interaction and accessibility, allowing users to receive alerts and data on their smartphones or tablets.
5. **Power Optimization**:
    - Further research into low-power components and energy-efficient algorithms could extend the battery life of portable applications, making the system more suitable for long-term deployment in remote or outdoor environments.
6. **Environmental Adaptability**:
    - Enhancing the system's ability to operate in various environmental conditions (e.g., extreme temperatures, humidity, or noisy environments) could broaden its applicability in industrial and outdoor settings.
7. **Educational Kits**:
    - Creating educational kits that include the ultrasonic radar system could promote STEM education, providing students with hands-on experience in electronics, programming, and robotics.
8. **Commercial Applications**:
    - Exploring commercial applications in sectors such as agriculture (for crop monitoring), healthcare (for patient monitoring), and security (for intruder detection) could lead to new business opportunities and innovations.

REFERENCES

When compiling a list of references for an Arduino-based ultrasonic radar system project, it is important to include a mix of books, articles, websites, and documentation that provide foundational knowledge, technical details, and practical guidance. Below is a sample list of references that could be relevant to such a project:

1. **Books:**

- Monk, Simon. *Programming Arduino: Getting Started with Sketches*. McGraw-Hill Education, 2016.
- Margolis, David. *Arduino Cookbook: Recipes to Begin, Expand, and Enhance Your Projects*. O'Reilly Media, 2011.
  Banzi, Massimo, and Michael Shiloh. *Getting Started with Arduino: The Open Source Electronics Prototyping Platform*. Maker Media, 2014.

2. Articles and Research Papers:
   - K. S. Rajasekaran, et al. "Design and Implementation of Ultrasonic Radar System for Object Detection." *International Journal of Engineering Research and Applications*, vol. 3, no. 5, 2013, pp. 123-128.
   - A. S. K. S. Kumar, et al. "Ultrasonic Sensor Based Distance Measurement System." *International Journal of Advanced Research in Computer Science and Software Engineering*, vol. 5, no. 4, 2015, pp. 123-126.
3. Websites and Online Resources:
   - Arduino Official Website. "Arduino Documentation." https://www.arduino.cc/en/Reference/HomePage.
   - SparkFun Electronics. "Ultrasonic Range Finder - HCsSR04." https://www.sparkfun.com/products/15569.
   - Adafruit Learning System. "Using the HC-SR04 Ultrasonic Distance Sensor." https://learn.adafruit.com/ultrasonic-range-finder-hc-sr04.
4. Datasheets:
   - HC-SR04 Ultrasonic Sensor Datasheet.
     https://www.electronicwings.com/public/images/product/HCSR04-Datasheet.pdf.
   - Arduino Uno Rev3 Datasheet.
     https://www.arduino.cc/en/uploads/Main/Arduino_Uno_Rev3schematic.pdf.
5. Forums and Community Contributions:
   - Arduino Forum. "Ultrasonic Sensor Projects." https://forum.arduino.cc/c/projects/ultrasonic-sensor-projects.
   - Instructables. "Arduino Ultrasonic Distance Sensor." https://www.instructables.com/Arduino-Ultrasonic-DistanceSensor/.